

Driver Glance Behavior Modeling Based on Semi-Supervised Clustering and Piecewise Aggregate Representation

Jianling Huang, Yan Long , and Xiaohua Zhao

Abstract—Glance behavior is significant because whether and how the driver is scanning and observing the driving scene is closely related to driving safety. This paper aims to improve the accuracy of glance behavior modeling and realize the spatiotemporal representation and visualization of glance behavior. Forty subjects were recruited to perform a freeway driving task using a driving simulator. The vehicle data were collected by the simulator. Drivers' gaze points were collected by an eye tracker. The prior knowledge on gaze points obtained through a statistical analysis were provided for K-means (KM) to form a semi-supervised K-means (SSKM), which classifies gaze points into different fixation zones. The classification results were compared with the results of KM. Furthermore, a clustering center-based piecewise aggregate representation (CCPAR) was proposed to characterize glance behavior. Maneuvers identification was taken as a case to evaluate the proposed method. The k-nearest neighbour (KNN) based on the similarity of CCPAR identified driving maneuvers into lane-keeping, left lane change, and right lane change. The identification results were compared with the results of the Hidden Markov Model (HMM). The average classification accuracies of KM and SSKM were 55.28% and 94.75%, respectively. The accuracies of maneuvers identified by CCPAR-KNN and by HMM were 87.50% and 85.83%, respectively. The results indicate that SSKM and CCPAR are feasible for glance behavior modeling. SSKM eliminates the randomness of initial cluster center selection and improves the accuracy of gaze points classification. CCPAR is intuitive and convenient to describe and visualize the spatiotemporal characteristics of glance behavior.

Index Terms—Glance behavior, semi-supervised clustering, piecewise aggregate representation, two-dimensional dynamic time warping.

I. INTRODUCTION

VISION is the main source of information for drivers, for whom information is collected through visual input.

Manuscript received 25 May 2020; revised 7 December 2020 and 6 April 2021; accepted 17 April 2021. Date of publication 25 May 2021; date of current version 8 July 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61876011. The Associate Editor for this article was M. Brackstone. (Corresponding author: Xiaohua Zhao.)

This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

Jianling Huang is with the College of Metropolitan Transportation, Beijing University of Technology, Beijing 100124, China, and also with the Beijing Transportation Information Center, Beijing 100073, China (e-mail: huangjianling@bjttw.gov.cn).

Yan Long and Xiaohua Zhao are with the College of Metropolitan Transportation, Beijing University of Technology, Beijing 100124, China (e-mail: longyan@emails.bjut.edu.cn; zhaoxiaohua@bjut.edu.cn).

Digital Object Identifier 10.1109/TITS.2021.3080322

Research based on eye movement data has become a popular approach for measuring drivers' visual perception. Glance consists of continuous fixations on an area of interest (AOI) and the transition between different AOIs [1], [2]. Drivers' glance behavior is a critical representation in understanding drivers' information gathering processes and visual attention and allocation mechanism. Glance behavior is significant because whether and how the driver is scanning and observing the driving scene is closely related to driving safety. During driving, the driver needs to scan different areas constantly to maintain awareness of the driving scene. Failure of observing proper locations may lead to a misjudgment of important information, increasing crash risks [3]. During car following, drivers' glances off the forward roadway for a period may cause a rear-end collision. During lane changing, failure of appropriate location observation, such as the mirror, is an important cause of crashes and near-crashes. Analysis of glance behavior is a useful way to assess crash risks and explore the cause of crashes, providing suggestions for collision avoidance. Glance behavior is an important cue to identify distracted driving. Glance behavior indexes, such as glance duration and glance location, have been used as valuable indicators in measuring and understanding distracted driving. Thus far, driver glance behavior has become a major research focus of human factors in car-following [4], lane-changing [5], the design and application of the eco-driving support system [6], distracted driving [7], fatigued driving, crashes and near-crashes [8], and second task [9].

To characterize glance behavior, the first step is to capture the driver's gaze points. The development of eye trackers, such as Tobii, SMI and Eyelink, makes it easier to capture eye movement data [10]. Generally, there are lots of gaze points in a driving task. A point-by-point analysis is time-consuming. If the gaze points are correctly divided into corresponding zones, we can analyze which zones drivers observe during driving and in what order of observation is. Setting fixation zones will help model and analyze the driver's glance behavior. Accordingly, we first review previous studies on how gaze points were classified into different fixation zones, and then review the researches on characterizing glance behavior.

A. Gaze Points Classification

Fixation is a series of raw gaze points. The fixation zone refers to the area of interest (AOI), i.e. at which the driver

is looking, such as windshields and side mirrors. Reasonable and effective division of driver's fixation zones is useful to quantify driver's visual behaviors. However, there is no standard for the number and division of fixation zones. Different researchers adopted different division methods. Fitch *et al.* [11] divided drivers' visual zones into eight zones: the front, front right, front left, left mirror, right mirror, rearview mirror, right window, and left window. Rangesh *et al.* [12] and Vora *et al.* [13] defined seven fixation zones: forward, right mirror, left mirror, rearview, speedometer, radio, and eyes closed. Olsen *et al.* [14] divided the visual zone into eleven fixation zones: forward, left window, left mirror, left blind spot, rearview mirror, right window, right mirror, right blind spot, instrument cluster, indeterminate, and other interior. Martin *et al.* [15] defined eight fixation zones: front, left, far left, right, front right, rear view, speedometer, and center stack.

Gaze points classification is the process of assigning gaze points to a fixation zone according to its position. Currently, the methods of gaze points classification can be divided into two types: manual classification and automatic classification. Manual classification requires human efforts to annotate all the gaze points manually. An annotator analyzes the position of the gaze point and categorizes it accordingly. Some researchers [16]–[18] used manual classification to annotate gaze points. Manual classification requires substantial manual intervention, making it an expensive and time-consuming process. Automatic classification was introduced to overcome this limitation. Automatic classification uses machine learning to automatically classify gaze points at a faster speed. Many researchers [10], [19], [20] have focused on the method of classifying gaze points into a fixation zone. K-means clustering algorithm, the most commonly used machine learning method, has its own disadvantages when used in gaze points classification. 1) The K-means specifies the same K value, namely, the number of fixation zones, which is not appropriate across all drivers and driving tasks. In the same driving task, some drivers look at the rear-view mirror but some do not. The number of fixation zones for drivers who look at the rear-view mirror and for those who do not are not the same. If K is set as a specific fixed value, the accuracy of classifying gaze points into fixation zones will consequently be low. 2) The K-means leads easily to misclassification because of the random selection of initial cluster centers. Different clustering results are obtained each time, and the accuracy is difficult to guarantee. Other unsupervised clustering algorithms, such as DBSCAN, have similar problems. Although DBSCAN does not need to specify the number of clusters, it must specify two other parameters: Eps (the radius of the neighborhood) and MinPts (the minimum number of neighbors). Therefore, its shortcomings are similar to K-means when used in gaze point classification. To overcome these shortcomings, this paper proposed the seed neighborhoods to construct seed sets, which help form a semi-supervised cluster algorithm to classify gaze points into different fixation zones. First, the experimenter selected different K values for each driver according to a heatmap and histogram of gaze points and the division of fixation zones. In other words, the number of fixation zones

of each driver was set manually. Second, the driver's visual field was divided into several bins, and the number of gaze points falling in each bin was counted. The bin with the largest density in each fixation zone was acted as prior knowledge to construct the seed neighborhood of the zone. Gaze points belonging to the seed neighborhood were used to construct the seed set. Third, the seed sets helped label the remaining gaze points.

B. Characterizing Glance Behavior

Characterizing glance behavior describes the process of moving eyes. Table I reflects the indicators and methods used for modeling glance behavior. Glance duration, glance position, and glance frequency are commonly used indicators in glance behavior analysis. After the indicators are obtained, the next step is to analyze them. In Table I, it can be seen that the current methods of analyzing glance behavior are traditional statistical methods (descriptive statistics, Friedman test, regression analysis), Markovian processes and hidden Markov models (HMM). It is well known that HMM is sensitive to the initial parameters. Descriptive statistics is used to summarize and describe the glance features, such as the number of glances and glance durations. It is difficult to characterize spatiotemporal characteristics and patterns of eye movement with descriptive statistics. The spatiotemporal characteristics are useful to analyze driver-vehicle interaction and operation. For example, a gaze points series is overlaid on the series of lane position so that a direct time comparison can be made to analyze the glance behavior during lane change. Accordingly, this work employs piecewise aggregation as a key method for describing and visualizing the spatiotemporal characteristics of glance behavior.

The objective of this paper is two-fold: 1) to present a semi-supervised K-means clustering to improve the accuracy of gaze points classification, and 2) to propose a clustering center-based piecewise aggregation method for describing and visualizing spatiotemporal characteristics of glance behavior. The rest of the paper is organized as follows: Section 2 presents the experimental design, which includes participants, apparatus, scenario, and process; Section 3 presents the methodology; Section 4 presents the results and discussions; and Section 5 summarizes the conclusions and proposes the future work.

II. EXPERIMENTAL DESIGN

A. Apparatus

The experiment was performed with an AutoSim driving simulator located in Beijing University of Technology. The simulator was made from a Toyota car. The steering wheel, safety belt and brake are exactly the same as the real car, as shown in Fig. 1. The driving simulator software is made up of three parts: Evariste (creating experimental scenarios), Simword (controlling the scenarios) and scanner (collecting data). Its horizontal and vertical field of view are 130° and 40° , respectively. A head-mounted eye-tracking system SMI was used to collect eye movement data. The gaze tracking range

TABLE I
STUDIES ON DRIVER GLANCE BEHAVIOR

Research Study	Task/ behavior	Indicators /Features	Method	Apparatus
Lee et al., 2004 [5]	Lane change	Eye glance position The number of eye glances Total glance time Mean glance time	Statistics Link probabilities	Camera
Li et al., 2019 [6]	Eco-safe driving	Average saccade duration Average glance duration Glance frequency	Friedman test Post-hoc test (Wilcoxon type)	FACELAB eye tracker
Zhang et al., 2016 [21]	Overtaking maneuvers	Gaze duration and frequency Saccade duration, frequency	Descriptive statistics	SMI eye tracker
Bakhita et al., 2019 [7]	Distracted driving Behavior	Glance durations Number of glances The distraction level index	Descriptive statistics Mixed-effects modeling	Camera
Tijerina et al., 2004 [4]	Car following	Glance locations Glance durations	Statistics Regression analysis	Camera
Olsen et al., 2004 [14]	Lane change	Glance location Number of glances Glance duration	Statistics Markovian process	Camera
Martin et al., 2018 [15]	Lane change	Gaze accumulation, Glance duration Glance frequency Glance location	Multivariate normal distribution	Camera
Birrell & Fowkes, 2014 [22]	Eco-driving Distracted driving	Glance frequency Glance duration Glance transitions	Statistics Probabilities	Camera
Yuan et al., 2011 [20]	Driving in free flow state	Glance location	Markov chain	EyeLink II eye tracker
Yuan et al., 2018 [18]	predicting driver's eyes-off-road	Glance duration Glance location	Hidden Markov Model (HMM)	Dikablis eye tracker
Lee et al., 2016 [23]	The second task (cell phone conversation)	Glance location	Hidden Markov Model (HMM) Random Forest	Camera

was 80° horizontally and 60° vertically. The tracking distance was over 40 cm.

B. Participants

Forty subjects were recruited from social personnel and college students after an advertisement was released. Three subjects conducted a preliminary experiment to check and test the feasibility of the experimental scheme. Thirty-seven subjects participated in the main experiment. Among them, 22 were males (age: 39±11 years, driving age: 15±10 years) and 15 were females (age: 31±10 years, driving age: 8±6 years). The ratio of male and female subjects is as close as possible to that of male and female drivers in China. The attributes of participants were shown in Table II. Drugs and alcoholic beverages were prohibited during the experiment. All subjects received a reward of no less than 400 RMB at the end of the experiment. All subjects signed an informed consent before the trial.

C. Scenario

The driving scenario presents a 46 km freeway that consists of straight and curved sections. It is a bidirectional

TABLE II
THE ATTRIBUTES OF PARTICIPANTS

Attributes		No. (%)
Gender	Male	22 (59.46%)
	Female	15 (40.54%)
Age (years)	18-35	19 (51.35%)
	36-45	8 (21.62%)
	46-55	10 (27.03%)
Driving age (years)	<11	19 (51.35%)
	11-20	11 (29.73%)
	>20	7 (18.92%)
Highest educational level	Middle school	2 (5.41%)
	High school	17 (45.95%)
	Bachelor or Master degree	18 (48.65%)
Professional driver	Yes	17 (45.95%)
	No	20 (54.05%)

four-lane freeway and the width of the road cross section is 26 m, the lane 3.75 m, and the green belt 2 m. The lane boundaries are marked clearly with white lines,



Fig. 1. The car simulator.



Fig. 2. The scenario.

as shown in Fig. 2. The minimum and maximum speed limits are 80 and 120 km/h, respectively. The speed limit signs are placed alongside the road. The vehicles travel in a free flow state. The drivers are allowed to change lanes freely at any place. The surrounding facilities, vehicles and speed limits are the only restrictions.

D. Experimental Process

Subjects were instructed to have at least 8 hours of sleep before the scheduled tasks. They were picked up by a researcher in the morning one hour before the start of the experiment. Subjects were instructed on the operation of the simulator and the experimental tasks. Then, the subject performed a test driving lasting about 5mins in adaptation to the system. After the test driving, subjects were asked to complete a questionnaire on their subjective feelings of the experiments and evaluations of the simulator. Subsequently, the real experiment was immediately performed.

E. Data Collection

The vehicle data included steering angle, speed, acceleration, lateral position (LP), gas pedal, and brake pedal. These data were collected at a frequency of 20 Hz. Eye movement data were recorded at a frequency of 60 Hz. To match the data of the driving simulator, the data of the eye tracker were downsampled.

Our experiment mainly studies on the eye movement process in lane changing process. There are no secondary activities. This paper takes lane change identification as an application case (example) to verify the proposed method. LP, defined as the distance between the center of the car and center road marking, is an important index in lane changing process (as shown in Fig. 3). We annotated the maneuvers

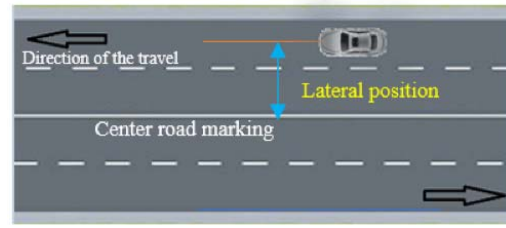


Fig. 3. Lateral position.

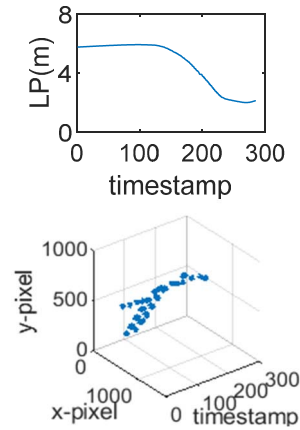


Fig. 4. LP and eye movement data of a left lane change.

and extracted the LP and eye movement data from about 3 s before the initial moment of lane change until the end of it. The interval is determined as three seconds because the critical decision-making period for lane change lasts for about three seconds [5], [24]. The total samples of LP and eye movement data of forty left lane changes, forty right lane changes, and forty lane-keepings were obtained. As an example, LP and eye movement data of a left lane change are shown in Fig. 4.

III. METHODOLOGY

A. Gaze Point Classification

In driving tasks, eye movement was captured by the eye tracker that produced a series of gaze points. A gaze location in the driver's visual field was represented by a gaze point that has horizontal and vertical coordinates (x , y) with respect to time. The prior knowledge from the statistical analysis on gaze points provides the K-means with the supervision information, which help form a semi-supervised K-means to classify gaze points into different fixation zones. The output of the algorithm is the zone to which each gaze point belongs. The main idea of the semi-supervised K-means is illustrated in Fig. 5.

1) *Division of the Fixation Zones*: There are few criteria for defining or dividing the fixation zone in literature. As stated in previous sections, different people have different methods of division. In this study, the valid eye movement data of 34 drivers were obtained (the rest 3 drivers' data were invalid due to vertigo or eye tracker slipping). After statistical analysis, it was found that there were three combinations of fixation zones, as shown in Table III. Thirty-one drivers (91.18%) looked at five zones during the driving process. One

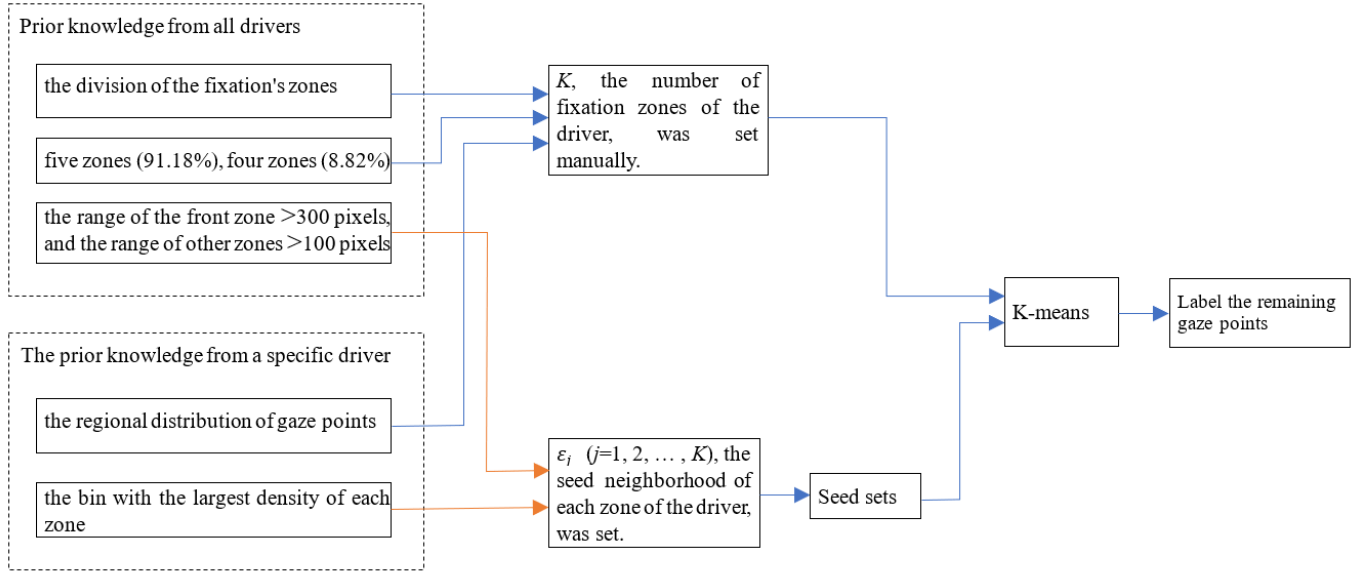


Fig. 5. The semi-supervised K-means.

TABLE III
THREE COMBINATIONS OF THE FIXATION ZONES

	front	left mirror	control panel	right mirror	rearview mirror	The number of drivers looked at the combination
Combination 1	✓	✓	✓	✓	✓	31 (91.18%)
Combination 2	✓	✓	✓	✓		1 (2.94%)
Combination 3	✓	✓	✓		✓	2 (5.88%)
The number of drivers looked at the zone	34 (100%)	34 (100%)	34 (100%)	32 (94.12%)	33 (97.06%)	

driver (2.94%) did not look at the rear-view mirror and two drivers (5.88%) did not look at the right mirror. According to the statistics, we specified five fixation zones: front, left mirror, control panel, right mirror, and rearview mirror, as shown in Fig. 6. The division of the fixation zones, which is the first prior knowledge, gives a general spatial layout. At the same time, the statistics also showed that for thirty-one drivers (91.18%), the number of fixation zones is five; and for three drivers (8.82%) the number is four, which is the second prior knowledge, as shown in Fig. 5. It is also found that the range of the front zone of the drivers is greater than 300 pixels, and that of other zones is greater than 100 pixels, which is the third prior knowledge.

Let FZ denote the set of five fixation zones.

$$\begin{aligned}
 FZ &= \{\text{front, left mirror, control panel, right mirror, rearview mirror}\} \\
 &= \{fz_1, fz_2, fz_3, fz_4, fz_5\}
 \end{aligned} \quad (1)$$

2) *Building a Histogram and Heatmap*: Gaze points over time during a driving task form a time series. In other words, a sequence of gaze points was output from the eye tracker. The three-sigma rule was used for outlier detection. The mean and



Fig. 6. The division of the fixation zones.

standard deviation of the horizontal and vertical coordinates of gaze points were computed. Points outside of ± 3 SD of the mean were identified as outliers. The outlier was replaced with the mean of the data before and after it. Gaze points after denoising were denoted as:

$$P = \begin{bmatrix} x(1), x(2), \dots, x(t), \dots, x(n) \\ y(1), y(2), \dots, y(t), \dots, y(n) \end{bmatrix} \quad (2)$$

where, x and y indicate the horizontal and vertical coordinates of the gaze point, respectively. t indicates the timestamp of the gaze point. n is the length of the gaze points series.

The x -direction, i.e., the horizontal direction of the driver's visual field, was divided into 30 bins, and the y -direction, i.e., the vertical direction, was also divided into 30 bins. The number of gaze points that fall in each of the 30×30 bins was counted. Accordingly, bin frequency $F(i, j)$ at bin (i, j) relative to gaze points at bin (i, j) was obtained. A histogram and a heatmap were then created based on bin frequency. The histogram shows the number of gaze points in bins. The heatmap better distinguishes different levels of gaze density. For a specific driver, the histogram and the heatmap of his gaze points provide us with the regional distribution and the spatial layout of gaze points, which is the fourth prior knowledge. The histogram can provide us with the bin with the largest density

in each fixation zone, which is the fifth prior knowledge (for details please refer to Section III.B.2). In short, the heatmap and the histogram of gaze points of the driver give us two aspects of prior knowledge, the spatial layout of gaze points and the bin with the largest density in each fixation zone of the driver.

3) *Classifying Gaze Points Into Different Fixation Zones Based on the Semi-Supervised K-Means*: Classifying gaze points into one of fixation zones is fundamental to exploring drivers' glance behavior. After statistical analysis, we obtained five aspects of prior knowledge, which play a crucial role in classifying gaze points. First, we determined the number of fixation zones of each driver, namely, the number of clusters, according to the division of the fixation zones (the first prior knowledge), the number of fixation zones of 5 or 4 (the second prior knowledge), and the regional distribution of gaze points (the fourth prior knowledge). Second, we found the centroid of the points in the bin with the largest density of each zone according to the fifth prior knowledge. The centroid of each zone was taken as the center. The radii were set according to the third prior knowledge. In this way, the seed neighborhoods were constructed. Gaze points belonging to the seed neighborhoods were used to construct the seed sets. We provided the determined number of fixation zones and the labeled seed sets for K-means to form the semi-supervised K-means. The seed sets have two functions. One is to initialize the cluster centers. The other is that the class labels of gaze points in the seed sets are already known. Labeled points were directly classified into their known clusters rather than changing the class labels during the iteration [25]–[27].

Briefly, the semi-supervised K-means has two improvements compared with K-means: 1) different values for K were chosen according to the heatmap of gaze points of each driver instead of applying a fixed K value to all drivers and drive tasks as K-means. 2) the seed neighborhood of each zone was proposed through statistical analysis. The points in the seed neighborhood were regarded as the seed set, which eliminates the randomness of initial cluster center selection and improves the accuracy of clustering. The algorithm is as follows.

B. Glance Behavior Modeling

Characterizing glance behavior describes the process of driver moving his/her eyes across time. Accordingly, the idea and method of time series can be introduced to model glance behavior. There are many different methods for representing and analyzing time series, such as piecewise linear transformation and piecewise aggregate representation [28], [29]. Based on the characteristics of eye movement data, this paper proposes a new method, clustering center-based piecewise aggregate representation (CCPAR), to represent and analyze a time series of gaze points. By applying CCPAR, researchers divide a time series of gaze points into q segments and utilize the cluster center to represent its segment, which focuses on reducing the dimension of data and mining patterns in gaze point time series. There are two main steps. The first step is segmentation that is to segment the original time series of gaze points; the second step is aggregation, where each segment is

Algorithm 1 Semi-Supervised K-Means Clustering

input: Set of gaze points $P = \begin{bmatrix} x(1), x(2), \dots, x(n) \\ y(1), y(2), \dots, y(n) \end{bmatrix}$

output: K partitions $\{Z_j\}_{j=1}^K$ of P and clustering centers $\mu_j (j = 1, 2, \dots, K)$.

method:

Step 1: the number of fixation zones K was set manually according to the heatmap and the division of the fixation zones.

Step 2: find the centroid $O_j (j = 1, 2, \dots, K)$ of the points in the bin with the largest density of each zone.

Step 3: for each zone, take the centroid O_j as the center and R_j as the radius to construct the seed neighborhood ε_j .

Step 4: construct seed set S_j for each zone

```

 $S_j = \emptyset$ 
for  $j = 1, 2, \dots, K$ 
  for all  $p \in P$ 
    if  $p \in \varepsilon_j$ 
       $S_j = S_j \cup \{p\}$ 
    end
  end
end

```

Step 5: the seed set $S = \bigcup_{j=1}^K S_j$ was manually labeled.

Step 6: calculate the initial cluster centers

```

for  $j = 1, 2, \dots, K$ 
   $\mu_j = \frac{1}{|S_j|} \sum_{p \in S_j} p$ 
end

```

Step 7: assign labeled points in the seed set to their known clusters

```

 $Z_j = \emptyset$ 
for  $j = 1, 2, \dots, K$ 
  for all  $p \in S_j$ 
     $Z_j = Z_j \cup \{p\}$ 
  end
end

```

Step 8: assign each unlabeled point to a new cluster

```

for  $p_i \in (P - S)$ 
   $d_{ij} = \|p_i - \mu_j\|^2$ 
   $r = \arg \min_{j \in \{1, 2, \dots, K\}} d_{ij}$ 
   $Z_r = Z_r \cup \{p_i\}$ 
end

```

Step 9: update the cluster centers

```

 $\mu_j = \frac{1}{|Z_j|} \sum_{p \in Z_j} p$ 

```

Step 10: repeat steps 8 and 9 until the algorithm converges.

represented by the clustering center of that segment. Details are as follows.

1) *Segmentation*: After the semi-supervised K-means was conducted, each gaze point was assigned to its corresponding category, that is, each gaze point has its own class label, namely, the fixation zone. The output of semi-supervised K-means clustering $\{Z_i\}_{i=1}^K$ was transformed into the fixation zone where each point is presented in chronological order.

A time series of fixation zones was accordingly formed, as shown in Equation (3)

$$G = [z(p_1), z(p_2), \dots, z(p_n)] \quad (3)$$

where, $z(p_i)$ is the class label of the gaze point p_i and indicates the fixation zone to which the gaze point p_i belongs.

The clustering centers of fixation zones are:

$$\Psi = \begin{Bmatrix} \mu_1^x & \mu_2^x & \dots & \mu_K^x \\ \mu_1^y & \mu_2^y & \dots & \mu_K^y \end{Bmatrix} \quad (4)$$

Segmentation manipulates the time series of gaze points into partitions that have similar gaze characteristics. A time series of gaze point segmentation facilitates the discovery of glance trends or glance patterns. For the time series of fixation zone G , the continuous sequence points in a fixation zone are assigned in one segment. For example, a time series $G = [fz_1fz_1fz_1fz_2fz_2fz_2fz_4fz_4fz_4fz_1fz_1fz_1fz_1fz_1]$ is a collection of sequential fixation zones of the gaze point $p_i (i = 1, 2, \dots, 15)$. G was divided into four segments $[fz_1fz_1fz_1]$, $[fz_2fz_2fz_2]$, $[fz_4fz_4fz_4]$, $[fz_1fz_1fz_1fz_1fz_1]$ by the boundaries $[1, 4, 7, 10]$. Segmentation of the time series of the fixation zone helps to find the boundaries. The algorithm is as follows.

Algorithm 2 Time-Series Segmentation

input: Time series of fixation zone

$$G = [z(p_1), z(p_2), \dots, z(p_n)]$$

output: The boundary set B

method:

```

Step 1: initialize  $B = \{1\}$ ,  $g = z(p_1)$ ,
Step 2: segment
  for  $i = 1$  to  $|G|$ 
    If  $g \neq z(p_i)$ 
       $B = B \cup \{i\}$ 
       $g = z(p_i)$ 
    end
  end
end

```

After time-series segmentation, the length of each segment was obtained. The length of each segment indicates the time spent on watching the corresponding zone. Mourant's study [24] showed that the typical value of a fixation is about 0.20 to 0.35 s. Meng's study [30] showed that the average fixation time is 0.15 to 0.23 s. The data were analyzed at a frequency of 20 Hz in the paper. Accordingly, the segment with a length of 3 sampling intervals or less (fixation time of 0.15 s or less) was too short for the driver to obtain meaningful information and was ignored. In other words, if the length of a segment was equal to or less than a threshold Υ (Υ was set to 3 sampling intervals in this paper), the segment was ignored and merged into its previous segment. The fixation zone where the segment belonged would be merged as a part into where its previous segment belonged. The above processing is equivalent to the smooth filtering of a time series of fixation zones. When Υ equals to 0, it indicates

Algorithm 3 Smooth Filtering

input: Time series of fixation zone

$G = [z(p_1), z(p_2), \dots, z(p_n)]$, the boundary set B

output: The updated G and B

method:

```

for all  $b \in B$ 
  if  $b_{i+1} - b_i \leq \Upsilon$ 
     $B = B - \{b_i\}$ 
     $z(p_{b_i}) = z(p_{b_{i-1}})$ 
     $Center(z(p_{b_i})) = Center(z(p_{b_{i-1}}))$ 
    if  $z(p_{b_{i-1}}) == z(p_{b_{i+1}})$ 
       $B = B - \{b_{i+1}\}$ 
    end
  end
end
end

```

that no smooth filtering was performed. The smooth filtering algorithm of a time series of the fixation zone is as follows.

After smooth filtering of a time series of the fixation zone was performed, the boundary set $B = [b_1, b_2, \dots, b_q] b_i \in \{1, 2, \dots, n\}$ was obtained. q segments $G_1, G_2, \dots, G_q$ were separated from G by the boundary set $B = [b_1, b_2, \dots, b_q]$, so that $b_1 = 1$. b_i is the left startpoint of the i th segment. The length of the i th segment can be calculated as $(b_{i+1} - b_i)$. In particular, the last segment, i.e., the length of the q th segment, is $(n - b_q + 1)$.

2) *Clustering Center-Based Piecewise Aggregate Representation*: After the segmentation of a time series of gaze points comes the next step, aggregation representation. Two parameters are needed to aggregate for G . The first parameter records the zone to which the segment belongs, and the second records the start point of the segment. Given a time series of fixation zones G , a CCPAR representation is as follows:

Representation I

$$\begin{aligned} \hat{G} &= [G_1, G_2, \dots, G_q] \\ &= [< Zone(s_1), b_1 >, < Zone(s_2), b_2 >, \\ &\quad \dots < Zone(s_q), b_q >] \end{aligned} \quad (5)$$

$Zone(s_i)$ is a function that obtains the zone to which segment i belongs

$$\begin{aligned} Zone(s_i) &= z(p_{b_i}) = fz_j \\ i &= 1, 2, \dots, q \quad j \in \{1, 2, \dots, K\} \end{aligned} \quad (6)$$

For a time series of gaze points P , the first parameter records the clustering center of the gaze point of the segment, and the second parameter records the start point of the segment.

$$\begin{aligned} \hat{P} &= [P_1, P_2, \dots, P_q] \\ &= [< Center(s_1), b_1 >, < Center(s_2), b_2 >, \\ &\quad \dots < Center(s_q), b_q >] \end{aligned} \quad (7)$$

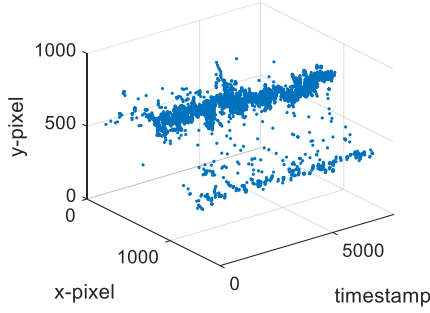


Fig. 7. The time series of gaze points.

$Center(s_i)$ is a function which obtains the clustering center of the i th segment

$$Center(s_i) = Center(fz_j) = \begin{bmatrix} \mu_j^x \\ \mu_j^y \\ \mu_j^t \end{bmatrix} \quad i = 1, 2, \dots, q \quad j \in \{1, 2, \dots, K\} \quad (8)$$

Representation I can be transferred into the second representation by joining the same fixation zone into a single set.

Representation II

$$\begin{aligned} \hat{G} &= [\langle fz_1, S_1 \rangle, \langle fz_2, S_2 \rangle, \dots, \langle fz_m, S_m \rangle] \quad m \leq K \\ S_1 &= \cup \{s_i\} \quad Zone(s_i) = fz_1 \quad \text{for } i = 1 \text{ to } q \\ S_2 &= \cup \{s_i\} \quad Zone(s_i) = fz_2 \quad \text{for } i = 1 \text{ to } q \\ &\dots\dots\dots \\ S_m &= \cup \{s_i\} \quad Zone(s_i) = fz_m \quad \text{for } i = 1 \text{ to } q \\ s_i &= \{b_i, b_i + 1, \dots, b_{i+1} - 1\} \quad i = 1, 2, \dots, q - 1 \\ s_q &= \{b_q, b_q + 1, \dots, n\} \\ \hat{P} &= [\langle Center(fz_1), S_1 \rangle, \langle Center(fz_2), S_2 \rangle, \dots, \langle Center(fz_m), S_m \rangle] \quad m \leq K \\ S_1 &= \cup \{s_i\} \quad Zone(s_i) = fz_1 \quad \text{for } i = 1 \text{ to } q \\ S_2 &= \cup \{s_i\} \quad Zone(s_i) = fz_2 \quad \text{for } i = 1 \text{ to } q \\ &\dots\dots\dots \\ S_m &= \cup \{s_i\} \quad Zone(s_i) = fz_m \quad \text{for } i = 1 \text{ to } q \\ s_i &= \{b_i, b_i + 1, \dots, b_{i+1} - 1\} \quad i = 1, 2, \dots, q - 1 \\ s_q &= \{b_q, b_q + 1, \dots, n\} \end{aligned} \quad (9)$$

An example is given to illustrate the CCPAR. In a driving task, the eye tracker captures the gaze location and outputs the result in form of the horizontal and vertical coordinates for 385.2 s. A time series of gaze points P with a length of 7,704 was formed as follows:

$$P = \begin{bmatrix} 811 & 815 & 813 & 816 \\ 761 & 780 & 770 & 785 & \dots \end{bmatrix} \quad |P| = 7704$$

The time series of gaze points over time is shown in Fig. 7. It is very time-consuming to manually assign these 7,704 gaze points to one of the fixation zones. And now we would explain our method step by step as below.

The (x -pixel, y -pixel) perspective of Fig. 7 is shown in Fig. (8-1). First, the x -direction was divided into 30 bins, and the y -direction, also into 30 bins. The number of gaze

points that fall in each of the 30×30 bins was counted. Accordingly, a heatmap and a histogram were built as shown in Fig. (8-2) and (8-3). The number of fixation zones K was manually set as 5 according to the heatmap and the division of fixation zones described in Section III.A.1. The bin with the largest density of each zone was then found according to the histogram, as shown in Fig. (8-3) and (8-4). The centroid of the bin with the largest density of each zone was found and taken as the center of the seed neighborhood of each zone. As stated above, it is found that the range of the front zone is greater than 300 pixels, and that of other zones is greater than 100 pixels. Accordingly, R_1 was set as 150 pixels and R_2, R_3, R_4, R_5 were set as 50 pixels. The seed neighborhood ε_j ($j = 1, 2, \dots, 5$) were constructed as shown in Fig. (8-5). The points in the seed neighborhoods formed a seed set and were labeled manually shown in Fig. (8-6). Then, the seed set helped label the remaining gaze points. K partitions $\{Z_j\}_{j=1}^5$ of P and clustering centers were obtained. The results divided by the semi-supervised K -means are shown in Fig. (8-7).

Second, the result of semi-supervised clustering $\{Z_j\}_{j=1}^5$ was transformed into fixation zones to which each gaze point belonged in chronological order. A time series of fixation zones G was accordingly formed.

$$\begin{aligned} G &= [z(p_1), z(p_2), \dots, z(p_{7704})] \\ &= [\underbrace{fz_1, fz_1, fz_1, \dots, fz_1}_{46}, fz_3, fz_3, fz_3, fz_4, fz_4, fz_4, fz_4, \dots] \\ |G| &= 7704 \\ FZ &= \{fz_1, fz_2, fz_3, fz_4, fz_5\} \\ \Psi &= \begin{bmatrix} \mu_1^x & \mu_2^x & \mu_3^x & \mu_4^x & \mu_5^x \\ \mu_1^y & \mu_2^y & \mu_3^y & \mu_4^y & \mu_5^y \end{bmatrix} \\ &= \begin{bmatrix} 825 & 217 & 1263 & 1436 & 1002 \\ 731 & 544 & 348 & 543 & 832 \end{bmatrix} \end{aligned}$$

Third, time-series segmentation was conducted. The continuous sequence points in a fixation zone are assigned in one segment. The time series of fixation zones G was separated into 173 segments, as follows.

$$\begin{aligned} G_1 &= [z(p_1), z(p_2), \dots, z(p_{46})] \\ &= [fz_1, fz_1, fz_1, \dots, fz_1] \\ G_2 &= [z(p_{47}), z(p_{48}), z(p_{49})] = [fz_3, fz_3, fz_3] \\ G_3 &= [z(p_{50}), z(p_{51}), z(p_{52}), z(p_{53}), z(p_{54})] \\ &= [fz_4, fz_4, fz_4, fz_4, fz_4] \\ &\dots\dots\dots \end{aligned}$$

The boundary set was obtained.

$$\begin{aligned} B &= \{1 \ 47 \ 50 \ 55 \ 56 \ 59 \ 261 \ 270 \ 422 \ 424 \ 427 \ 428 \ 434 \\ &\quad 511 \ 517 \ 518 \ 694 \ 699 \ 708 \ 744 \ 757 \ 773 \ 907 \ 915 \\ &\quad 1027 \ 1028 \ 1031 \ 1033 \ 1035 \ 1037 \ 1146 \ 1147 \ 1148 \\ &\quad 1165 \ 1178 \dots 7704\} \end{aligned}$$

The length of each segment is shown in Fig. (9-1), and it can be seen that there are some burrs with a length no more

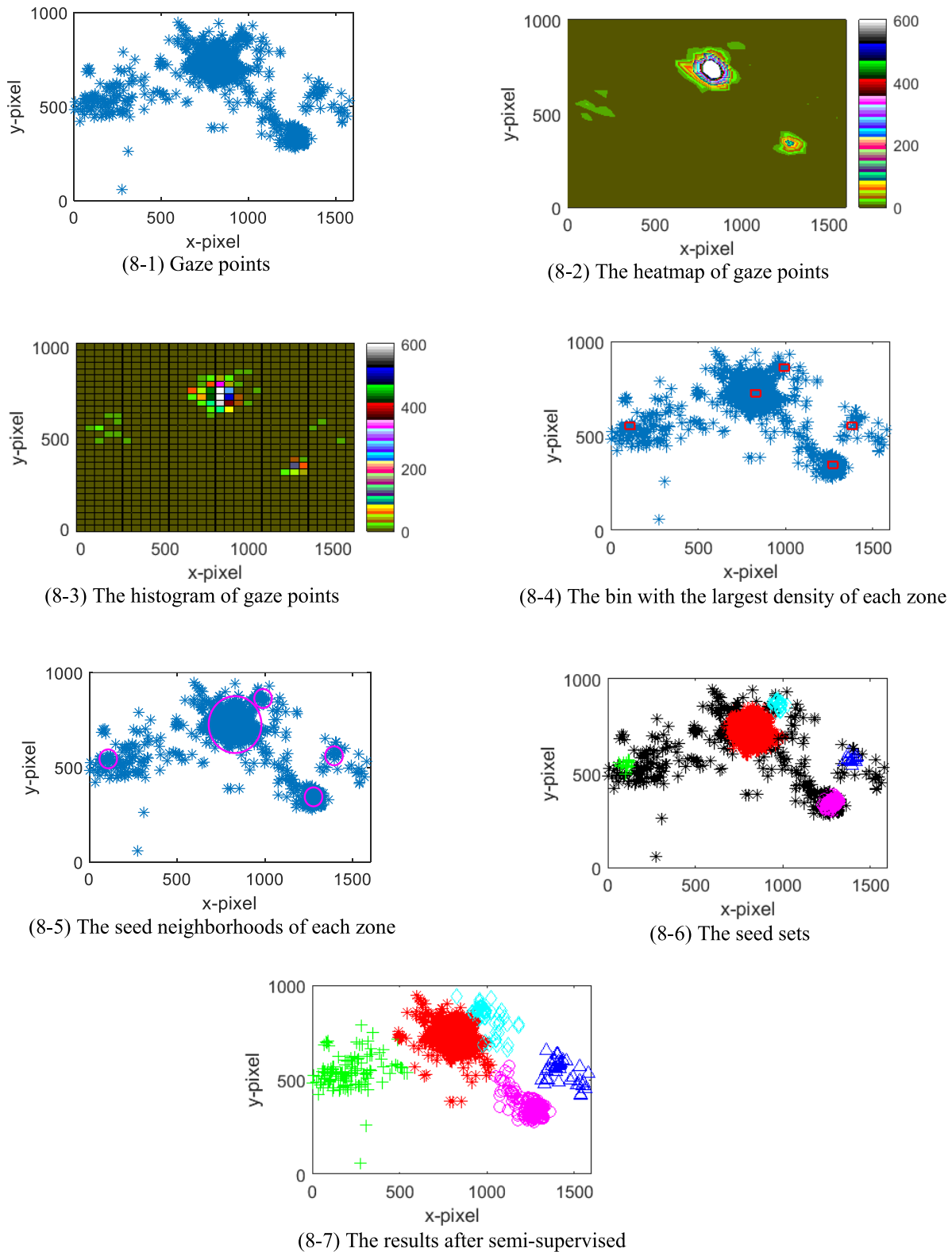


Fig. 8. The process of classifying gaze points.

than 3 sampling intervals. A smooth filtering algorithm was performed, and the filtered results are shown in Fig. (9-2). The updated boundary set B was also obtained. The time series of gaze points before and after filtering are shown

in Fig. (10-1) and Fig. (10-2), respectively. Comparisons of the gaze points before and after filtering from the perspective of (timestamp, x -pixel) are shown in Fig. (10-3) and Fig. (10-4).

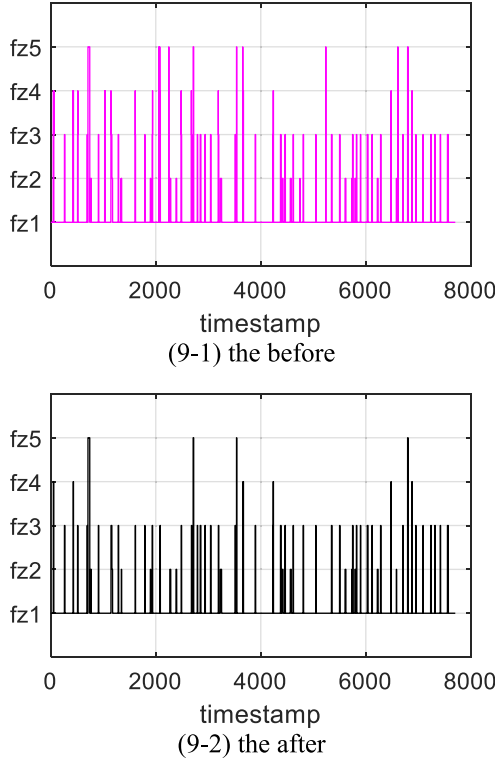


Fig. 9. Comparison of the segmentation before and after smooth filtering.

Finally, the updated boundary set obtained is as follows

$$B = \{1 \ 50 \ 59 \ 261 \ 270 \ 428 \ 434 \ 511 \ 518 \ 694 \ 699 \ 709 \\ 744 \ 757 \ 773 \ 907 \ 915 \ 1148 \ 1165 \ 1178 \dots 7704\}$$

According to equation (7), the CCPAR representation I of the time series of gaze points is as follows:

$$\hat{P} = [P_1, P_2, \dots, P_q]$$

$$P_1 = \langle \text{Center}(s_1), b_1 \rangle = \langle (825, 731), 1 \rangle$$

$$P_2 = \langle \text{Center}(s_2), b_2 \rangle = \langle (1436, 543), 50 \rangle$$

.....

$$P_i = \langle \text{Center}(s_i), b_i \rangle$$

.....

$$P_{127} = \langle \text{Center}(s_{127}), b_{127} \rangle = \langle (825, 731), 7572 \rangle$$

According to equation (10), the CCPAR representation I of the time-series of gaze points is as follows

$$\hat{P} = [\langle \text{Center}(fz_1), S_1 \rangle, \langle \text{Center}(fz_2), S_2 \rangle, \\ \langle \text{Center}(fz_3), S_3 \rangle, \langle \text{Center}(fz_4), S_4 \rangle, \\ \langle \text{Center}(fz_5), S_5 \rangle]$$

$$= [\langle (825, 731), \{s_1, s_3, s_5, s_7, s_9, s_{11}, s_{13}, s_{15}, \\ s_{17}, \dots, s_{127}\} \rangle,$$

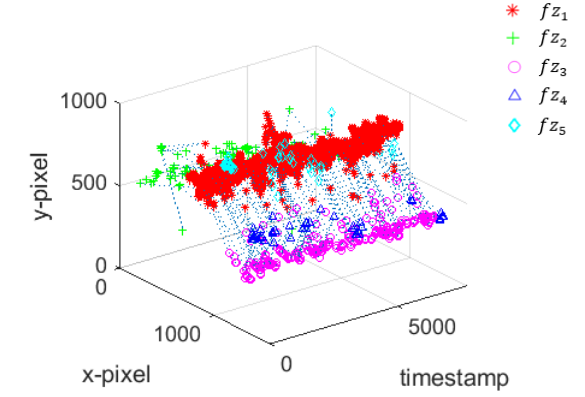
$$\langle (217, 544), \{s_{14}, s_{19}, s_{23}, \dots, s_{102}\} \rangle,$$

$$\langle (1263, 348), \{s_4, s_8, s_{10}, \dots, s_{124}\} \rangle,$$

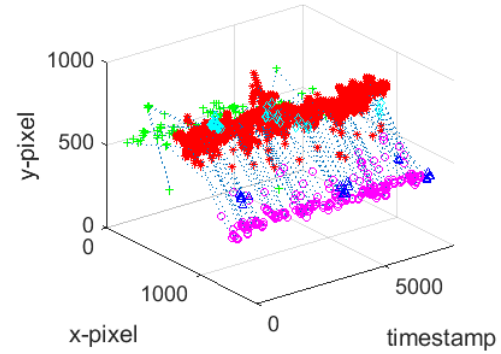
$$\langle (1436, 543), \{s_2, s_6, s_{61}, \dots, s_{112}\} \rangle,$$

$$\langle (1002, 832), \{s_{12}, s_{43}, s_{59}, s_{110}\} \rangle]$$

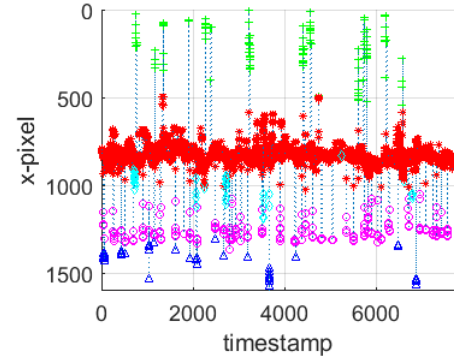
$$s_1 = \{1, 2, 3, \dots, 49\}$$



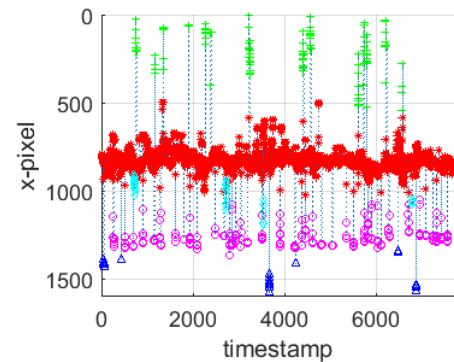
(10-1) The time series of gaze points before filtering



(10-2) The time series of gaze points after filtering



(10-3) The before in view of (timestamp, x-pixel)



(10-4) The after in view of (timestamp, x-pixel)

Fig. 10. Comparison of gaze points before and after filtering.

$$s_2 = \{50, 51, 52, \dots, 58\}$$

.....

$$s_{127} = \{7572, 7573, \dots, 7704\}$$

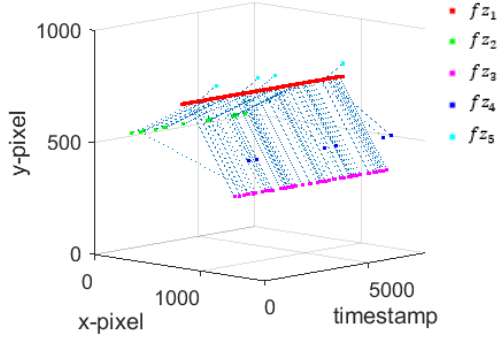


Fig. 11. The CCPAR of gaze points.

TABLE IV
DESCRIPTIVE STATISTICS

	The number of glances	Total glance time(s)	Mean glance time(s)
fz_1	63	351.2	5.6
fz_2	14	10.1	0.7
fz_3	40	18.2	0.5
fz_4	6	2.6	0.4
fz_5	4	3.1	0.8

A graphical representation of the CCPAR is shown in Fig. 11. CCPAR represents a time series of gaze points using the boundary set and clustering centers. It reduces the storage space and transforms the time series shown in Fig. 7 into another time series shown in Fig. 11. We compare the CCPAR proposed by this paper with descriptive statistics. The results of the descriptive statistics are shown in Table IV. Descriptive statistics help to summarize the number of glances and glance durations of each zone. Compared with CCPAR shown in Fig. 11, it is difficult to characterize the spatiotemporal characteristics of glance behavior with descriptive statistics. CCPAR models better the driver glance behavioral characteristics across time and space. Of course, traditional indicators, such as glance duration, glance position, and glance frequency, are easily obtained with CCPAR.

C. Lane Change Identification

1) *2D-DTW*: Left or right lane change can be identified by analyzing the similarity of CCPARs of glance behavior. The most commonly used measure of similarity is Euclidean distance. However, Euclidean distance has some shortcomings. First, the Euclidean distance requires two time series with the same length. Second, Euclidean distance does not allow contraction or expansion of the time series along the time axis. DTW (dynamic time warping) overcomes the limitations by distorting and shifting the series in the time dimension [31]–[33].

Let $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_m\}$ be two time series with the length of n and m , respectively. An n -by- m matrix C is constructed to represent the distance $d(i, j)$ between the two points x_i and y_j .

$$d(i, j) = \|x_i - y_j\| \quad i \in [1 : n], \quad j \in [1 : m] \quad (11)$$

The matrix C is generally called the local cost matrix. Let $W = \{w_1, w_2, \dots, w_p, \dots, w_P\}$ ($\max(m, n) \leq P < m + n - 1$) denote the warping path, where P is the length of the warp path. The p th element of the path is $w_p = (i, j)$. The warping path is generally subject to the following criteria [31]–[33].

Boundary condition:

$$w_1 = (1, 1) \text{ and } w_P = (n, m).$$

Monotonicity condition:

$$w_p = (i, j) w_{p+1} = (i', j') \quad i \leq i' \leq i + 1, \quad j \leq j' \leq j + 1$$

Step size condition:

$$(w_{p+1} - w_p) \in \{(1, 0), (0, 1), (1, 1)\}, \quad p \in [1 : P - 1]$$

The *distance* or *cost* $d_W(X, Y)$ of a warping path W between X and Y is defined as

$$d_W(X, Y) = \sum_{p=1}^P d(w_{pi}, w_{pj}) \quad (12)$$

where, $d(w_{pi}, w_{pj})$ is the *distance* or *cost* between the two point indexes (one from time series X and one from Y) of the p th element of the warp path W .

Generally, there are many warping paths satisfying the above criteria. However, we are interested in the optimal warp path, which is the minimal cost path between the two series. The *DTW distance* between the time series X and Y is defined as the cost of the optimal warp path.

$$DTW(X, Y) = \min\{d_W(X, Y) | W \text{ is a warping path}\} \quad (13)$$

The above describes the DTW for two one-dimensional time series. However, the horizontal and vertical coordinates of the gaze point are obtained at each timestamp, thus creating a two-dimensional time series. In this case, we need a generalized DTW, which can calculate the distance between two two-dimensional time series. In this paper, we employed the method proposed by Gillian [34] and Choi [35] to extend the standard DTW to two-dimensional DTW (2D-DTW).

Let $P_A = \begin{bmatrix} x_A(1), x_A(2), \dots, x_A(n) \\ y_A(1), y_A(2), \dots, y_A(n) \end{bmatrix}$ and $P_B = \begin{bmatrix} x_B(1), x_B(2), \dots, x_B(m) \\ y_B(1), y_B(2), \dots, y_B(m) \end{bmatrix}$ be two two-dimensional time series of gaze points with the length of n and m , respectively. The distance matrix C is as follows,

$$d(i, j) = \sqrt{(x_A(i) - x_B(j))^2 + (y_A(i) - y_B(j))^2} \quad i \in [1 : n], \quad j \in [1 : m] \quad (14)$$

The standard DTW was extended to 2D-DTW using the above distance matrix, and the 2D-DTW was calculated. The 2D-DTW is identical to the standard DTW with the exception of the calculation of $d(i, j)$.

It is necessary to centralize the time series of gaze points before computing the 2D-DTW between them. Because drivers possess different heights and habits, the cluster center of the fixation zone of the front (fz_1) is different. To compare the similarities, it is necessary to shift them to a baseline. This is accomplished by panning the center points of fz_1 of each driver to the average position of the center points of fz_1 of all drivers.

2) *KNN Classifier*: K-nearest neighbors (KNN) is an important method in pattern recognition and classification. KNN compares the similarity between testing samples and training samples, and then classifies the testing samples according to the majority in the neighbors. A KNN classifier was used to classify maneuvers into three classes (left lane change, right lane change and lane-keeping) according to the similarity of CCPARs of glance behavior under different maneuvers. The CCPAR was used as a feature vector for classification. The 2D-DTW was used to measure the similarity. The algorithm was conducted on a Windows 10 machine with a 2.50 GHz CPU and 8.0 GB of RAM. The 5-fold cross-validation was used to test the model performance.

3) *HMM Classifier*: We compared our proposed CCPAR-KNN with the Hidden Markov Model (HMM). The number of hidden states (maneuvers) was three, which are left lane change, right lane change and lane-keeping. The sequences of glance duration in five zones, derived from descriptive statistics, were used as observation status. The initial parameters, such as the transition probability matrix and the initial probability distribution, were randomly selected. Baum-Welch algorithm was then used to estimate the final parameters based on the initial parameters. One HMM classifier is built for each class, so three HMM classifiers were built for left lane change, right lane change and lane-keeping. A test sequence was input to the three classifiers and then was classified to the class with the maximum likelihood probability. The toolbox provided by Kevin Murphy [36] was used to implement the HMM.

IV. RESULTS AND DISCUSSION

A. Results

1) *Gaze Points Classification Results*: The proposed semi-supervised K-means was used to classify the gaze points into different zones. The classification of gaze points collected from 34 drivers was completed. For each drive, the classification accuracy (ACC) of all gaze points was calculated. And the precision, recall and F1-score of gaze points of each zone were obtained. All zones were treated equally regardless of how often they appear in the gaze process. Accordingly, the Macro-Precision, Macro-Recall and Macro-F1 of gaze points of each driver were calculated as arithmetic mean of individual zone's precision, recall and F1-score, respectively. We compared the semi-supervised K-means with unsupervised K-means. Table V gives the comparison of the results.

2) *Lane Change Identification Results*: For KNN, there is no standard approach for finding the optimal value of “ K ”. In this paper, we detected the labels with K equal to 1, 3, 5, ..., 21. For each K value, the algorithm is executed for 10 times. Fig. 12 shows the average performance of the KNN classifier. From Fig. 12, it can be seen that the best value for K was 7, which achieved the best average accuracy, and then we used 7 as the final K value of the KNN classifier. We selected the result of the highest ACC in 10 times as the final identification result. The confusion matrix is shown in Fig. 13. From the confusion matrix,

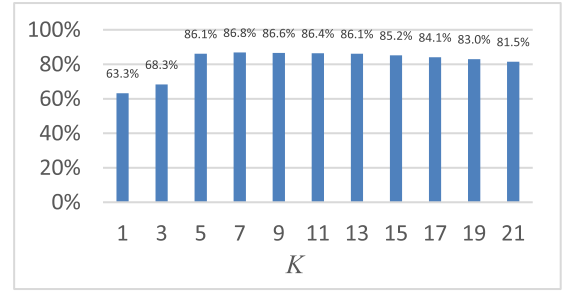


Fig. 12. The average accuracy for each K executed 10 times.

		Detected class			Total
		Left lane change	Right lane change	Lane keeping	
True class	Left lane change	36	1	3	40
	Right lane change	0	38	2	40
	Lane keeping	7	2	31	40
	total	43	41	36	120

Fig. 13. The confusion matrix of CCPAR-KNN.

		Detected class			Total
		Left lane change	Right lane change	Lane keeping	
True class	Left lane change	39	0	1	40
	Right lane change	0	39	1	40
	Lane keeping	11	4	25	40
	total	50	43	27	120

Fig. 14. The confusion matrix of HMM.

we calculated the precisions, recalls, and F1, as shown in Table VI. CCPAR-KNN had a total accuracy of 87.50%.

For HMM, the initial parameters were randomly selected for 100 times, so classifiers were constructed correspondingly for 100 times. Among the identification results of the 100 times, the highest ACC is 85.83% and the lowest ACC is 70.00%. We choose to compare the highest result with the result of CCPAR-KNN. The confusion matrix is shown in Fig. 14. From the confusion matrix, we calculated the precisions, recalls, and F1, as shown in Table VI. The classifier had a total accuracy of 85.83%. ACC, Macro-Precision, Macro-Recall and Macro-F1 of CCPAR-KNN and HMM are shown in Fig. 15.

B. Discussion

Table V proves the effectiveness of the gaze point classification based on the semi-supervised K-means clustering. Gaze point classification has a crucial role in modeling glance

TABLE V
GAZE POINTS CLASSIFICATION RESULTS

	ACC				Macro-F1				Macro-Precision				Macro-Recall			
	Mean	Max	Min	Std	Mean	Max	Min	Std	Mean	Max	Min	Std	Mean	Max	Min	Std
K-means	55.28%	97.95%	14.23%	20.25%	62.90%	93.87%	28.13%	14.67%	60.42%	94.99%	33.00%	13.56%	74.01%	94.67%	53.01%	11.43%
Semi-supervised K-Means	94.75%	99.55%	78.88%	4.74%	76.79%	96.68%	51.25%	13.61%	73.56%	94.32%	47.27%	13.82%	93.28%	99.91%	69.83%	8.75%

TABLE VI
LANE CHANGE IDENTIFICATION RESULTS

method	Left lane change			Right lane change			Lane keeping		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
CCPAR-KNN	83.72%	90.00%	86.75%	92.68%	95.00%	93.83%	86.11%	77.50%	81.58%
HMM	78.00%	97.50%	86.67%	90.70%	97.50%	93.98%	92.59%	62.50%	74.63%

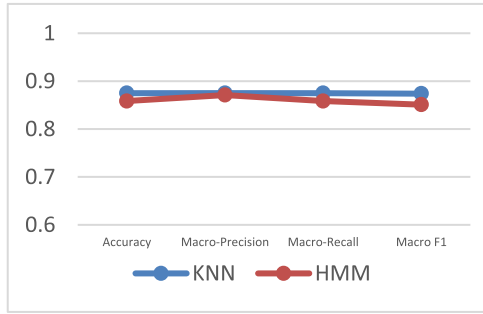


Fig. 15. The performance indexes.

behavior. Reasonable and effective gaze points classification is a prerequisite for glance behavior analysis. For K-means, different clustering results are obtained at each time based on the random selection of initial cluster centers. Fig. 16 shows two clustering results of gaze points collected from the same driver. Obviously, it is difficult to guarantee the accuracy of K-means. This paper proposed the semi-supervised method to classify gaze points. Different K values can be set for different drivers in various driving tasks through the heatmap and the histogram. The cluster centers were initialized by the seed sets rather than randomly selected. Accordingly, the semi-supervised method produces a more accurate classification of gaze points than the unsupervised method as shown in Table V. In addition, with this method, researchers only need to find the largest bin of each zone and construct the seed sets with the radius according to the prior knowledge. The rest work was left to the machine. After that, the clustering of all points was achieved. To classify the gaze points of a specific driver, though in multiple driving tasks, only two tasks need to be done manually: choosing K and finding the largest bin in each zone. Therefore, this method requires less manual intervention. In short, compared with unsupervised K-means, the proposed method improves the accuracy. Compared with manual classification, it saves time and cuts cost.

It should be noted that the semi-supervised clustering is based on the division of fixation zones, which depends on the research purpose. Different research purposes require different divisions of fixation zones. The selection of radius of the seed neighborhood is dependent on the research purpose and experiment designs. In this paper, R_1 was set as 150 pixels

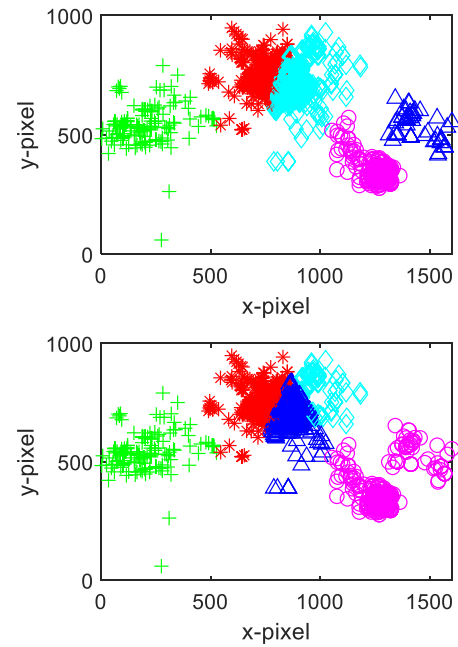


Fig. 16. Different results produced by K-means.

and R_2, R_3, R_4, R_5 were set as 50 pixels for analyzing the glance behavior of lane changing. For the glance behavior of distracted driving, fatigued driving, and second task, the selection of seed neighborhood radius needs to be further studied and determined according to the specific situation.

The CCPAR of the time series of gaze points is feasible and reliable. A graphical representation of the CCPAR is shown in Fig. 11. CCPAR transforms the time series shown in Fig. 7 into another time series shown in Fig. 11. The proposed method is an intuitive, simple, but efficient compression technology. Compared with the original time series of gaze points shown in Fig. 7, the CCPAR of gaze points not only reduces the storage space but also intuitively describes the driver's scanning path and dwell time. Compared with descriptive statistics, CCPAR better captures driver glance behavior at various times and characterizes unique glance behavioral patterns. For example, a CCPAR is overlaid on the series of lane position so that a direct temporal comparison can be made to analyze the glance behavior under lane

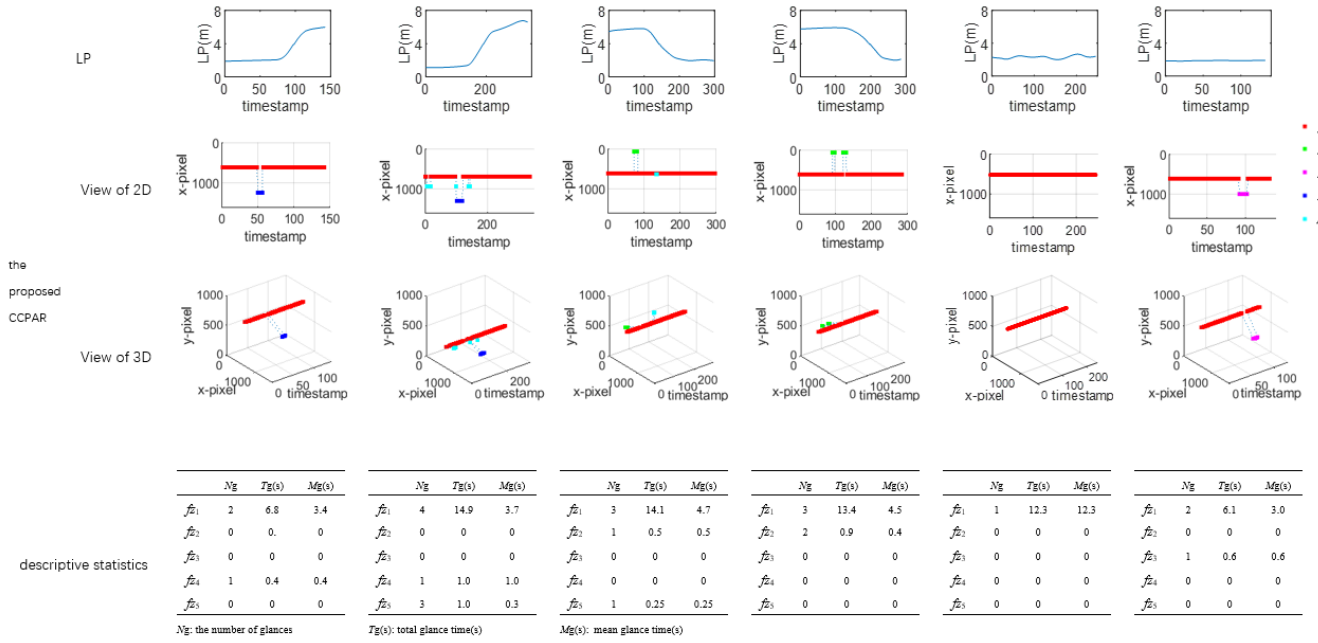


Fig. 17. The comparison between CCPAR and descriptive statistics.

change maneuvers as shown in Fig. 17. The spatiotemporal characteristics is shown in form of visualization, it is more helpful to understand and analyze the driver-vehicle interaction and operation. CCPAR can model the driver's glance pattern when maneuvers (right lane change, left lane change, and lane-keeping) are performed. Most drivers looked at the left mirror when he changed to the left lane and looked at the right mirror when he changed to the right lane. CCPAR can clearly characterize glance behavior, as shown in Fig. 17. This common glance characteristic of drivers is the inherent mechanism for successfully classifying maneuvers.

The benefits of using the cluster centroids of fixation zones in the CCPAR without using the label of fixation zones are as follows. 1) the cluster centroids provide the location information, which are useful for the application about the location, such as DTW algorithm. 2) the cluster centroids can be used to construct the 3-dimension scan path over time. 3) the cluster centroids reflect the individual differences of drivers. The cluster centroids of fixation zones are different among drivers. 4) the cluster centroids enable the visualization of glance behavior.

The high accuracy of lane change identification indicates that the proposed method is feasible. Table VI and Fig. 15 show that the accuracy of CCPAR-KNN outperforms that of HMM. CCPAR has observable/understandable features and patterns to describe the driver's eye movement. 2D-DTW can measure the similarity of CCPAR across time and space. By combining CCPAR with 2D-DTW, a white-box model is provided, which is transparent and explainable, for understanding the relationship between driver's eye movement and maneuvers. The maneuver identification model has not only observable input-output but also the clarity of internal work. It is transparent and its predictions are traceable by humans. Some drivers did not look at the left mirror when

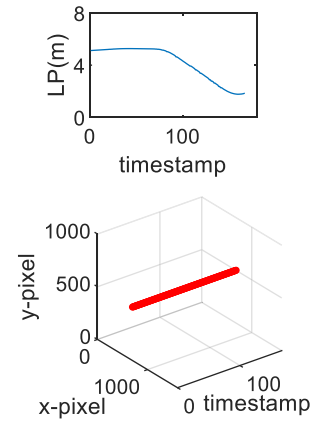


Fig. 18. Left lane change without looking at the left mirror.

performing a left lane change, as shown in Fig. 18. Left lane changes without looking at the left mirror were incorrectly classified as lane-keeping because most drivers looked at the left mirror when performing a left lane change. The reason for incorrect classification is clear.

V. CONCLUSION

Driver glance behavior is fundamental to the driver's visual attention mechanism. This paper focused on the two key issues of modeling glance behavior, i.e., identifying the fixation zone each gaze point belongs to, and then characterizing glance behavior.

To address the first issue, a semi-supervised cluster algorithm was introduced to determine which zone each point belonged to. The number of fixation zones was manually determined according to a heatmap of gaze points and the division of fixation zones. The seed neighborhood of each zone was created and gaze points belonging to the seed neighborhoods

were acted as supervision information to construct seed sets. The seed set helped label the remaining gaze points. Briefly, the improvement of the semi-supervised K-means compared with K-means clustering is twofold.

1) Instead of applying a fixed K value to all drivers and drive tasks as K-means, our proposed method chooses the different K values according to the heatmap of gaze points of each driver.

2) The seed neighborhood of each zone was proposed through statistical analysis. The points in the seed neighborhood were regarded as the seed set to form a semi-supervised K-means, which eliminates the randomness of initial cluster center selection and improves the accuracy of clustering.

To address the second issue, we proposed a piecewise aggregation method to describe and visualize glance behavior across time and space. An application of maneuver identification was performed to evaluate the proposed method. A KNN classifier took the CCPAR under the maneuvers as the input, analyzed the similarity, and then assigned the corresponding label. The results showed that the proposed method is feasible. The advantages of CCPAR are as follows:

1) CCPAR is an intuitive, simple, but efficient compression technology. Only two parameters are needed to aggregate.

2) CCPAR characterizes and visualizes spatiotemporal characteristics of glance behavior effectively and depicts glance patterns intuitively.

3) CCPAR has a clear physical meaning. Combining with 2D-DTW, it is a white-box operation used to classify glance patterns and understand glance behavior.

One limitation is that the proposed method in this paper is offline postprocessing. Real-time processing will be studied later. Additionally, the feasibility and efficiency of the proposed method in identifying distracted or fatigued driving, analyzing glance behavior during car-following, and mining glance patterns in crashes or near-crashes need to be further studied. The universality and generalization of the proposed method require to be further evaluated.

REFERENCES

- [1] P. Green, "Where do drivers look while driving (and for how long)," in *Human Factors in Traffic Safety*, 2nd ed. Tucson, AZ, USA: Lawyers Judges Publishing, 2007, pp. 57–82.
- [2] M. Bergstrand. (2008). Automatic analysis of eye tracker data from a driving simulator. Linköping University, Sweden. [Online]. Available: <http://www.diva-portal.org/smash/record.jsf?pid=diva2:636736>
- [3] Y. Wang, S. Bao, W. Du, Z. Ye, and J. R. Sayer, "Examining drivers' eye glance patterns during distracted driving: Insights from scanning randomness and glance transition matrix," *J. Saf. Res.*, vol. 63, pp. 149–155, Dec. 2017, doi: [10.1016/j.jsr.2017.10.006](https://doi.org/10.1016/j.jsr.2017.10.006).
- [4] L. Tijerina, F. S. Barickman, and E. N. Mazzae, "Driver eye glance behavior during car following," US DOT NHTSA, Washington, DC, USA, Tech. Rep. DOT HS 809 723, Apr. 2004.
- [5] S. E. Lee, E. C. B. Olsen, and W. W. Wierwille, "A comprehensive examination of naturalistic lane-changes," US DOT NHTSA, Washington, DC, USA, Tech. Rep. DOT HS 809 702, Mar. 2004.
- [6] X. Li, A. Vaezipour, A. Rakotonirainy, and S. Demmel, "Effects of an in-vehicle eco-safe driving system on drivers' glance behaviour," *Accident Anal. Prevention*, vol. 122, pp. 143–152, Jan. 2019.
- [7] P. R. Bakhit, O. A. Osman, B. Guo, and S. Ishak, "A distraction index for quantification of driver eye glance behavior: A study using SHRP2 NEST database," *Saf. Sci.*, vol. 119, pp. 106–111, Nov. 2019.
- [8] J. Bärghman, V. Lisovskaja, T. Victor, C. Flannagan, and M. Dozza, "How does glance behavior influence crash and injury risk? A 'what-if' counterfactual simulation using crashes and near-crashes from SHRP2," *Transp. Res. F, Traffic Psychol. Behaviour*, vol. 35, pp. 152–169, Nov. 2015.
- [9] D. L. Smith, J. Chang, R. Glassco, J. Foley, and D. Cohen, "Methodology for capturing driver eye glance behavior during in-vehicle secondary tasks," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1937, no. 1, pp. 61–65, Jan. 2005.
- [10] Q. Xu, T.-Y. Guo, F. Shao, and X.-J. Jiang, "Division of area of fixation interest for real vehicle driving tests," *Math. Problems Eng.*, vol. 2017, pp. 1–10, Jan. 2017, doi: [10.1155/2017/3674374](https://doi.org/10.1155/2017/3674374).
- [11] G. M. Fitch, S. E. Lee, S. Klauer, J. Hankey, J. Sudweeks, and T. Dingus, "Analysis of lane-change crashes and near-crashes," US DOT NHTSA, Washington, DC, USA, Tech. Rep. DOT HS 811 147, Jun. 2009.
- [12] A. Rangesh, B. Zhang, and M. M. Trivedi, "Driver gaze estimation in the real world: Overcoming the eyeglass challenge," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Oct. 2020, pp. 1054–1059.
- [13] S. Vora, A. Rangesh, and M. M. Trivedi, "Driver gaze zone estimation using convolutional neural networks: A general framework and ablative analysis," *IEEE Trans. Intell. Vehicles*, vol. 3, no. 3, pp. 254–265, Sep. 2018.
- [14] E. C. B. Olsen, S. E. Lee, and W. W. Wierwille, "Eye glance behavior during lane changes and straight-ahead driving," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1937, no. 1, pp. 44–50, Jan. 2005.
- [15] S. Martin, S. Vora, K. Yuen, and M. M. Trivedi, "Dynamics of driver's gaze: Explorations in behavior modeling and maneuver prediction," *IEEE Trans. Intell. Vehicles*, vol. 3, no. 2, pp. 141–150, Jun. 2018.
- [16] H. Z. Xu and Y. L. Pei, "A model of driver's fixation point shift during lane change and its parameter calibration," *J. Highway Transp. Res. Develop.*, vol. 27, no. 11, pp. 118–121, 2010.
- [17] O. Lappi, P. Rinkkala, and J. Pekkanen, "Systematic observation of an expert driver's gaze strategy—An on-road case study," *Frontiers Psychol.*, vol. 8, pp. 1–15, Apr. 2017, doi: [10.3389/fpsyg.2017.00620](https://doi.org/10.3389/fpsyg.2017.00620).
- [18] W. Yuan, Z. F. Liu, and R. Fu, "Predicting drivers' eyes-off-road duration in different driving scenarios," *Discrete Dyn. Nature Soc.*, vol. 2018, pp. 1–10, Jan. 2018, doi: [10.1155/2018/3481628](https://doi.org/10.1155/2018/3481628).
- [19] S.-W. Li *et al.*, "Driver fixation region division-oriented clustering method based on the density-based spatial clustering of applications with noise and the mathematical morphology clustering," *Adv. Mech. Eng.*, vol. 7, no. 10, pp. 1–11, 2015.
- [20] W. Yuan, R. Fu, Y. S. Guo, and C. L. Xue, "Analysis of driver's visual transition characters with Markov chain," in *Proc. Int. Conf. Softw. Eng. Multimedia Commun.*, Qingdao, China, 2011, pp. 294–298.
- [21] W. Zhang, J. Dai, Y. Pei, P. Li, Y. Yan, and X. Chen, "Drivers' visual search patterns during overtaking maneuvers on freeway," *Int. J. Environ. Res. Public Health*, vol. 13, no. 11, p. 1159, Nov. 2016.
- [22] S. A. Birrell and M. Fowkes, "Glance behaviours when using an in-vehicle smart driving aid: A real-world, on-road driving study," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 22, pp. 113–225, Jan. 2014.
- [23] J. Lee, M. Muñoz, L. Fridman, T. Victor, B. Reimer, and B. Mehler, "Investigating drivers' head and glance correspondence," 2016, *arXiv:1602.07324*. [Online]. Available: <http://arxiv.org/abs/1602.07324>, doi: [10.7717/peerj-cs.146](https://doi.org/10.7717/peerj-cs.146).
- [24] R. R. Moutant and R. J. Donohue, "Mirror sampling characteristics of drivers," SAE Tech. Paper 740964, 1974, doi: [10.4271/740964](https://doi.org/10.4271/740964).
- [25] X. Wang, C. Wang, and J. Shen, "Semi-supervised K -means clustering by optimizing initial cluster centers," in *Proc. Int. Conf. Web Inf. Syst. Mining*, 2011, pp. 178–187.
- [26] S. Basu, A. Banerjee, and R. Mooney, "Semi-supervised clustering by seeding," in *Proc. 19th Int. Conf. Mach. Learn. (ICML)*, 2002, pp. 19–26.
- [27] E. Bair, "Semi-supervised clustering methods," *Wiley Interdiscipl. Rev., Comput. Statist.*, vol. 5, no. 5, pp. 349–361, Sep. 2013.
- [28] J. Lin, E. Keogh, S. Lonardi, and B. Chiu, "A symbolic representation of time series, with implications for streaming algorithms," in *Proc. 8th ACM SIGMOD Workshop Res. Issues Data Mining Knowl. Discovery (DMKD)*, San Diego, CA, USA, 2003, pp. 2–11.
- [29] K. Chakrabarti, E. Keogh, S. Mehrotra, and M. Pazzani, "Locally adaptive dimensionality reduction for indexing large time series databases," *ACM Trans. Database Syst.*, vol. 27, no. 2, pp. 188–228, Jun. 2002.
- [30] N. Meng, "The analysis on driver's fixation process in different traffic environment road," Master's thesis, School Automobile, Chang'an Univ., Xi'an, China, 2009.

- [31] P. Senin, "Dynamic time warping algorithm review," Dept. Inf. Comput. Sci., Univ. Hawaii Manoa, Honolulu, HI, USA, Tech. Rep., Dec. 2018, pp. 1–23. [Online]. Available: https://seninp.github.io/assets/pubs/senin_dtw_litreview_2008.pdf
- [32] S. Salvador and P. Chan, "Toward accurate dynamic time warping in linear time and space," *Intell. Data Anal.*, vol. 11, no. 5, pp. 561–580, Oct. 2007.
- [33] E. J. Keogh and M. J. Pazzani, "Derivative dynamic time warping," in *Proc. SIAM Int. Conf. Data Mining*, Apr. 2001, pp. 1–11.
- [34] N. Gillian, R. B. Knapp, and S. O'Modhrain, "Recognition of multi-variate temporal musical gestures using N-dimensional dynamic time warping," in *Proc. 11th Int. Conf. New Interfaces Musical Expression*, 2011, pp. 337–342.
- [35] H.-R. Choi and T.-Y. Kim, "Directional dynamic time warping for gesture recognition," in *Proc. 2nd Int. Conf. Multimedia Syst. Signal Process.*, 2017, pp. 22–25.
- [36] K. Murphy. (1998). *Hidden Markov Model (HMM) Toolbox for MATLAB*. [Online]. Available: <http://www.cs.ubc.ca/~murphyk/Software/HMM/hmm.html>



Jianling Huang was born in Guangxi, China. She received the M.S. degree in applied mathematics from Tianjin University, Tianjin, China, and the Ph.D. degree in highway and railway engineering from Beijing Jiaotong University, Beijing, China.

She is currently a Professor with the College of Metropolitan Transportation, Beijing University of Technology, and the Director of the Beijing Transportation Information Center. Her research interests include intelligent transportation systems, transportation planning, and management.



Yan Long was born in Liaoning, China, in 1979. She is currently pursuing the Ph.D. degree with the College of Metropolitan Transportation, Beijing University of Technology. Her research interests include driving behavior, transportation planning, and management.



Xiaohua Zhao was born in Shanxi, China, in 1971. She received the Ph.D. degree in pattern recognition and artificial intelligence from the Beijing University of Technology.

She is currently a Professor with the College of Metropolitan Transportation, Beijing University of Technology. Her research interest includes driving behavior.